

8th International Conference on Machine Learning & Applications (CMLA 2026)

July 16 ~ 17, 2026, London, United Kingdom

<https://cmla2026.org/index>

Scope & Topics

The 8th International Conference on Machine Learning & Applications (CMLA 2026) provides a premier global forum for researchers, practitioners, and industry experts to share the latest advances in machine learning theory, methodologies, and real world applications. As machine learning continues to transform science, engineering, business, and society, CMLA 2026 aims to bring together a diverse community to exchange ideas, present innovative research, and explore emerging challenges and opportunities in the field.

The conference welcomes high quality contributions that demonstrate significant progress in foundational machine learning, cutting edge algorithms, large scale systems, and domain specific applications. Authors are invited to submit original research articles, case studies, survey papers, and industrial experiences that highlight meaningful advances in machine learning and its rapidly expanding ecosystem.

CMLA 2026 encourages submissions across a broad range of topics, including but not limited to the areas listed below. By fostering collaboration between academia, industry, and research institutions, the conference seeks to accelerate innovation, deepen scientific understanding, and support the development of next generation machine learning technologies.

Topics of interest include, but are not limited to, the following

Foundations of Machine Learning

- Statistical Learning Theory and Generalization
- Optimization for ML (Convex, Non Convex, Large Scale)
- Probabilistic Modeling, Bayesian Learning and Graphical Models
- Causal Inference, Causal ML and Counterfactual Reasoning
- Online Learning, Meta Learning and Continual Learning
- Multi Task Learning, Transfer Learning and Domain Adaptation
- Theory of Deep Learning and Emergent Behaviors

Deep Learning and Representation Learning

- Neural Network Architectures and Training Techniques
- Self Supervised Learning and Contrastive Learning
- Generative Models (GANs, Diffusion Models, VAEs)
- Diffusion Models for Images, Text, Time Series, Molecules and Graphs
- Foundation Models, LLMs, Vision Language Models and Multimodal Models
- Efficient Deep Learning (Pruning, Quantization, Distillation)
- Representation Learning for Structured, Temporal and Graph Data

Reinforcement Learning, Decision Making and Embodied AI

- Deep Reinforcement Learning and Policy Optimization
- Multi Agent RL, Game Theory and Coordination
- Offline RL, Safe RL and Risk Sensitive RL
- World Models, Embodied AI and Interactive Learning
- RL for Robotics, Control Systems and Real World Deployment
- Hierarchical RL and Skill Discovery
- Planning Augmented Models and Decision Transformers

Natural Language Processing, Speech and Multimodal AI

- Large Language Models and Instruction Tuned Models
- Retrieval Augmented Generation (RAG) and Knowledge Grounded Models
- Long Context Models, Memory Augmented Models and Tool Using LLMs
- Text Generation, Summarization and Dialogue Systems
- Speech Recognition, Speech Synthesis and Audio Language Models
- Vision Language Models, Video Language Models and Multimodal Fusion
- NLP for Low Resource Languages and Cross Lingual Learning

Computer Vision, Perception and Graphics

- Image Classification, Detection and Segmentation
- 3D Vision, Scene Understanding and SLAM
- Vision Transformers and Diffusion Based Vision Models
- Video Understanding, Action Recognition and Motion Prediction
- Generative Vision Models, Neural Rendering and 3D Generation
- Embodied Perception and Interactive Vision
- Vision Language Action Models for Robotics

Data Mining, Knowledge Discovery and Graph Learning

- Graph Neural Networks (GNNs) and Graph Representation Learning
- Knowledge Graphs, Reasoning and Neuro Symbolic AI
- Large Scale Data Mining and Pattern Discovery
- Time Series Forecasting, Anomaly Detection and Predictive Modeling
- Simulation Based ML and Synthetic Data Generation
- ML for Structured, Relational and Heterogeneous Data

Trustworthy, Explainable and Responsible AI

- Explainable AI (XAI) and Mechanistic Interpretability
- Fairness, Accountability, Transparency and Ethics in ML
- Robust ML, Adversarial Attacks and Defenses
- Jailbreak Resistant LLMs and Safety Evaluation
- Privacy Preserving ML (Differential Privacy, Federated Learning, Secure ML)
- Safety Critical ML and Reliability
- AI Governance, Risk Assessment and Policy Aligned ML

ML Systems, Hardware Acceleration and Efficient Computing

- Distributed and Parallel ML Systems
- Training and Inference Optimization for Foundation Models
- ML Compilers, Optimization and Deployment Frameworks
- Edge ML, TinyML and On Device Learning
- Edge Native Foundation Models and Distributed Inference
- Neuromorphic Computing and Brain Inspired ML
- Energy Efficient ML, Green AI and Carbon Aware ML Pipelines

Applied Machine Learning and Domain Specific ML

- Healthcare and Life Sciences
- Medical Imaging, Diagnostics and Clinical Decision Support
- Computational Biology, Genomics and Drug Discovery
- Digital Health, Wearables and Personalized Medicine
- ML for Neuroscience and Cognitive Modeling
- ML for Digital Therapeutics and Clinical Decision Automation

Science and Engineering

- ML for Physics, Chemistry, Materials Science and Climate Modeling
- Physics Informed ML and Scientific Machine Learning (SciML)
- Differentiable Physics, Neural Simulators and ML Accelerated Simulation
- ML for Robotics, Autonomous Systems and Control
- ML for Smart Cities, IoT and Cyber Physical Systems

Business, Finance and Social Systems

- ML for Finance, Risk Modeling and Fraud Detection
- Recommender Systems, Personalization and User Modeling
- Social Network Analysis and Computational Social Science
- ML for Policy Simulation and Societal Impact Modeling

Emerging Trends

- Agentic AI, Autonomous AI Systems and Multi Agent LLM Ecosystems
- Tool Using AI, Planning Augmented LLMs and Autonomous Agents
- Program Synthesis, AI for Code and ML Guided Theorem Proving
- Quantum Machine Learning and Quantum Inspired Algorithms
- AutoML, Neural Architecture Search (NAS) and Hyperparameter Optimization
- ML for Foundation Model Alignment, Safety and Governance
- ML for Autonomous Scientific Discovery and Robot Scientists
- ML for Synthetic Biology, Bio Inspired Algorithms and Living Systems

Paper Submission

Authors are invited to submit papers through the conference [Submission System](#) by **May 09, 2026**. Submissions must be original and should not have been published previously or be under consideration for publication while being evaluated for this conference. The proceedings of the conference will be published by [Computer Science Conference Proceedings](#) (H index 46) in [Computer Science & Information Technology \(CS & IT\)](#) series (Confirmed).

Selected papers from **CMLA 2026**, after further revisions, will be published in the special issues of the following journals.

- [Machine Learning and Applications: An International Journal \(MLAIJ\)](#)
- [International Journal of Artificial Intelligence & Applications \(IJAIA\)](#)
- [Advances in Vision Computing: An International Journal \(AVC\)](#)

Important Dates

- **Submission Deadline: May 09, 2026**
- Authors Notification: June 20, 2026
- Registration & Camera-Ready Paper Due: June 27, 2026

Contact Us

Here's where you can reach us: cmla@cmla2026.org (or) confer_cmla@yahoo.com

For more details, please visit: <https://cmla2026.org/index>

Paper Submission Link: <https://cseit2026.org/submission/index.php>